

TOPSIS MCDM Analysis Report

Fuzzy TOPSIS preserves triangular/trapezoidal uploaded values through normalization, weighting, fuzzy ideal distances, and closeness coefficients.

Study Summary

Alternatives/factors: 5. Criteria/factors: 7. Top result: Epsilon.

94 bundled numerical checks and 43 external published-example fixtures are included; TOPSIS, AHP, DEMATEL, VIKOR, COPRAS, WASPAS, SAW/WSM, WPM, MOORA, MOOSRA, MULTIMOORA, MABAC, CODAS, CoCoSo, ARAS, EDAS, PROMETHEE II, ELECTRE I, SMART, MAUT, SMARTER, Pugh Matrix, OCRA, ROV, GRA, MARCOS, MAIRCA, PSI, PIV, WISP, RAM, PROBID, SPROBID, LMAW, RIM, RAFSI, SPOTIS, B-SPOTIS, MARA, LoPM, AROMAN, COMET, and ERVD external-example fixtures are registered; TOPSIS, CRADIS, fuzzy DEMATEL, and the richer B-SPOTIS used-car published-example candidate are tracked separately; paper-by-paper comparisons for every crisp, fuzzy, and group-decision variant are not yet complete.

Certification status: not publication-certified for every variant. Only registered passing external fixtures count as published-example validation evidence for the selected method and variant.

Study Mode Evidence

Input type: 35 fuzzy cells detected. Fuzzy types: triangular, trapezoidal. Mode: Native fuzzy TOPSIS.

Respondent/group aggregation: No respondent matrix aggregation used. The uploaded/active decision matrix was treated as a single already aggregated dataset.

Expert aggregation: No DEMATEL expert matrix aggregation used. Only DEMATEL multiple-expert studies use expert direct-relation aggregation.

Fuzzy calculation convention: Native fuzzy TOPSIS. Native fuzzy methods report the selected fuzzy mode; centroid-only methods report defuzzified input handling.

Validation Evidence

Certification status: Not publication-certified for every variant. Treat internal checks as implementation evidence. Treat only registered passing external fixtures as published-example validation for the selected method and variant.

Selected method validation status: Validated with discrepancies tracked. 1 passing published-example fixture; 3 validation candidates tracked separately

Internal workflow and algorithm checks: Included. Registry integrity, method execution, group data, fuzzy input, zero-value handling, validation rules, templates, reports, and coverage summaries.

Bundled numerical benchmark checks: 94 checks. Known expected rankings and key coefficients for the built-in sample suites.

Selected method benchmark coverage: TOPSIS. Selected method has bundled numerical benchmark coverage

External published-example validation: In progress: 43 fixtures. TOPSIS, AHP, DEMATEL, VIKOR, COPRAS, WASPAS, SAW/WSM, WPM, MOORA, MOOSRA, MULTIMOORA, MABAC, CODAS, CoCoSo, ARAS, EDAS, PROMETHEE II, ELECTRE I, SMART, MAUT, SMARTER, Pugh Matrix, OCRA, ROV, GRA, MARCOS, MAIRCA, PSI, PIV, WISP, RAM, PROBID, SPROBID, LMAW, RIM, RAFSI, SPOTIS, B-SPOTIS, MARA, LoPM, AROMAN, COMET, and ERVD external-example fixtures are registered; TOPSIS, CRADIS, fuzzy DEMATEL, and the richer B-SPOTIS used-car published-example candidate are tracked separately; paper-by-paper comparisons for every crisp, fuzzy, and group-decision variant are not yet complete.

Selected method external validation: TOPSIS. 1 external published-example fixture registered for this selected method: crisp-vector-normalization-ahp-weights-hospital-supplier; 3 validation candidates also tracked.

External fixture: TOPSIS crisp-vector-normalization-ahp-weights-hospital-supplier. Journal of Healthcare Engineering 2019 hospital supplier TOPSIS example; DOI 10.1155/2019/5614892. TOPSIS vector normalization, AHP-derived manual weights, benefit/cost-aware ideal solutions, Euclidean separation distances, closeness coefficients, and final supplier ranking. Source: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6811789/>

External fixture: TOPSIS crisp-vector-normalization-manual-weights (discrepancy candidate). Central European Journal of Operations Research 2026 TOPSIS warehouse-storage example; DOI 10.1007/s10100-026-01038-6. Published TOPSIS matrix and weights reproduce the top alternative but not all closeness values or the middle ranking under documented vector-normalized TOPSIS settings. Source: <https://link.springer.com/article/10.1007/s10100-026-01038-6>

External fixture: TOPSIS crisp-vector-normalization-ahp-weights (discrepancy candidate). SpringerPlus 2016 AHP-TOPSIS ETL software-selection example; DOI 10.1186/s40064-016-1888-z. Published TOPSIS input and AHP-derived weights reproduce the top and bottom alternatives but not all closeness values or the middle ranking under documented vector-normalized TOPSIS settings. Source: <https://pmc.ncbi.nlm.nih.gov/articles/PMC4775722/>

External fixture: TOPSIS crisp-vector-normalization-ahp-weights-barge-service (discrepancy candidate). Asian Journal of Shipping and Logistics 2025 barge-service supplier TOPSIS example; DOI 10.1016/j.ajsl.2024.11.002. Published TOPSIS matrix, AHP weights, and closeness coefficients do not match standard vector-normalized TOPSIS; the normalized table appears to reuse a denominator across criteria with different column norms. Source: <https://doi.org/10.1016/j.ajsl.2024.11.002>

Calculation Steps

- 1. Compute vector-normalized decision matrix.
- 2. Apply criteria weights.
- 3. Determine positive and negative ideal solutions.
- 4. Calculate separation distances and closeness coefficient.

Analyzed Decision Matrix

Alternative	C1	C2	C3	C4	C5	C6	C7
Alpha	6	8.5	10	12.5	14	16.5	18
Beta	9	12.5	15	18.5	21	24.5	27
Gamma	12	16.5	20	24.5	28	32.5	36
Delta	15	20.5	25	30.5	35	40.5	45
Epsilon	18	24.5	30	36.5	42	48.5	54

Fuzzy Normalized Matrix

Alternative	C1	C2	C3	C4	C5	C6	C7
Alpha	(0.7143, 0.8333, 1) (0.2692, 0.3077, 0.3462, 0.3846) (0.8182, 0.9, 1) (0.2895, 0.3158, 0.3421, 0.3684) (0.3023, 0.3256, 0.3488) (0.3, 0.32, 0.34, 0.36) (0.8947, 0.9444, 1)						
Beta	(0.5, 0.5556, 0.625) (0.4231, 0.4615, 0.5, 0.5385) (0.5625, 0.6, 0.6429) (0.4474, 0.4737, 0.5, 0.5263) (0.4651, 0.4884, 0.5116) (0.46, 0.48, 0.5, 0.52) (0.6071, 0.6296, 0.6538)						
Gamma	(0.3846, 0.4167, 0.4545) (0.5769, 0.6154, 0.6538, 0.6923) (0.4286, 0.45, 0.4737) (0.6053, 0.6316, 0.6579, 0.6842) (0.6279, 0.6512, 0.6744) (0.62, 0.64, 0.66, 0.68) (0.4595, 0.4722, 0.4857)						
Delta	(0.3125, 0.3333, 0.3571) (0.7308, 0.7692, 0.8077, 0.8462) (0.3462, 0.36, 0.375) (0.7632, 0.7895, 0.8158, 0.8421) (0.7907, 0.814, 0.8372) (0.78, 0.8, 0.82, 0.84) (0.3696, 0.3778, 0.3864)						
Epsilon	(0.2632, 0.2778, 0.2941) (0.8846, 0.9231, 0.9615, 1) (0.2903, 0.3, 0.3103) (0.9211, 0.9474, 0.9737, 1) (0.9535, 0.9767, 1) (0.94, 0.96, 0.98, 1) (0.3091, 0.3148, 0.3208)						

Fuzzy Weighted Normalized Matrix

Alternative	C1	C2	C3	C4	C5	C6	C7
Alpha	(0.102, 0.119, 0.1429) (0.0385, 0.044, 0.0495, 0.0549) (0.1169, 0.1286, 0.1429) (0.0414, 0.0451, 0.0489, 0.0526) (0.0432, 0.0465, 0.0498) (0.0429, 0.0457, 0.0486, 0.0514) (0.1278, 0.1349, 0.1429)						
Beta	(0.0714, 0.0794, 0.0893) (0.0604, 0.0659, 0.0714, 0.0769) (0.0804, 0.0857, 0.0918) (0.0639, 0.0677, 0.0714, 0.0752) (0.0664, 0.0698, 0.0731) (0.0657, 0.0686, 0.0714, 0.0743) (0.0867, 0.0899, 0.0934)						
Gamma	(0.0549, 0.0595, 0.0649) (0.0824, 0.0879, 0.0934, 0.0989) (0.0612, 0.0643, 0.0677) (0.0865, 0.0902, 0.094, 0.0977) (0.0897, 0.093, 0.0963) (0.0886, 0.0914, 0.0943, 0.0971) (0.0656, 0.0675, 0.0694)						
Delta	(0.0446, 0.0476, 0.051) (0.1044, 0.1099, 0.1154, 0.1209) (0.0495, 0.0514, 0.0536) (0.109, 0.1128, 0.1165, 0.1203) (0.113, 0.1163, 0.1196) (0.1114, 0.1143, 0.1171, 0.12) (0.0528, 0.054, 0.0552)						
Epsilon	(0.0376, 0.0397, 0.042) (0.1264, 0.1319, 0.1374, 0.1429) (0.0415, 0.0429, 0.0443) (0.1316, 0.1353, 0.1391, 0.1429) (0.1362, 0.1395, 0.1429) (0.1343, 0.1371, 0.14, 0.1429) (0.0442, 0.045, 0.0458)						

Fuzzy Ideal Best and Worst

Type	C1	C2	C3	C4	C5	C6	C7
Fuzzy ideal best	(0.0376, 0.0376, 0.0376) (0.1429, 0.1429, 0.1429) (0.0415, 0.0415, 0.0415) (0.1429, 0.1429, 0.1429) (0.1429, 0.1429, 0.1429) (0.1429, 0.1429, 0.1429) (0.1429, 0.1429, 0.1429)						
Fuzzy ideal worst	(0.1429, 0.1429, 0.1429) (0.0385, 0.0385, 0.0385) (0.1429, 0.1429, 0.1429) (0.0414, 0.0414, 0.0414) (0.0432, 0.0432, 0.0432) (0.0429, 0.0429, 0.0429) (0.1429, 0.1429, 0.1429)						

Fuzzy TOPSIS Distances and Closeness

Alternative	D+	D-	Closeness
Alpha	0.6497	0.0812	0.111
Beta	0.4277	0.2866	0.4013
Gamma	0.2725	0.4405	0.6178
Delta	0.1439	0.5693	0.7982
Epsilon	0.0327	0.6854	0.9545

Applied Criteria Weights

Criterion	Name	Direction	Applied Weight
C1	Cost	cost	0.1429
C2	Environmental Management	benefit	0.1429
C3	CO2 Emissions	cost	0.1429
C4	Labor Practices	benefit	0.1429
C5	Innovation	benefit	0.1429
C6	Community Impact	benefit	0.1429
C7	Waste	cost	0.1429

Final Ranking

Rank	Alternative	Score
1	Epsilon	0.9545
2	Delta	0.7982
3	Gamma	0.6178
4	Beta	0.4013
5	Alpha	0.111

Sensitivity Analysis

Criterion	Name	Base Weight	Low Scenario	High Scenario	Top Alternative	Stability Note
C1	Cost	0.1429	0.1286	0.1571	Epsilon	Review with +/-10% weight perturbation
C2	Environmental Management	0.1429	0.1286	0.1571	Epsilon	Review with +/-10% weight perturbation
C3	CO2 Emissions	0.1429	0.1286	0.1571	Epsilon	Review with +/-10% weight perturbation
C4	Labor Practices	0.1429	0.1286	0.1571	Epsilon	Review with +/-10% weight perturbation
C5	Innovation	0.1429	0.1286	0.1571	Epsilon	Review with +/-10% weight perturbation
C6	Community Impact	0.1429	0.1286	0.1571	Epsilon	Review with +/-10% weight perturbation
C7	Waste	0.1429	0.1286	0.1571	Epsilon	Review with +/-10% weight perturbation

Figure data: Ranking Bar Chart

alternative	score	rank
Epsilon	0.9545	1
Delta	0.7982	2
Gamma	0.6178	3
Beta	0.4013	4
Alpha	0.111	5

Figure data: Criteria Weights

criterion	weight
C1	0.1429
C2	0.1429
C3	0.1429
C4	0.1429
C5	0.1429
C6	0.1429
C7	0.1429

Figure data: Decision Matrix Heatmap

alternative	criterion	value
Alpha	C1	6
Alpha	C2	8.5
Alpha	C3	10
Alpha	C4	12.5
Alpha	C5	14
Alpha	C6	16.5
Alpha	C7	18
Beta	C1	9
Beta	C2	12.5
Beta	C3	15
Beta	C4	18.5
Beta	C5	21

Beta | C6 | 24.5
Beta | C7 | 27
Gamma | C1 | 12
Gamma | C2 | 16.5
Gamma | C3 | 20
Gamma | C4 | 24.5
Gamma | C5 | 28
Gamma | C6 | 32.5
Gamma | C7 | 36
Delta | C1 | 15
Delta | C2 | 20.5
Delta | C3 | 25
Delta | C4 | 30.5
Delta | C5 | 35
Delta | C6 | 40.5
Delta | C7 | 45
Epsilon | C1 | 18
Epsilon | C2 | 24.5
Epsilon | C3 | 30
Epsilon | C4 | 36.5
Epsilon | C5 | 42
Epsilon | C6 | 48.5
Epsilon | C7 | 54

Figure data: Weight Perturbation Band

criterion	name	baseWeight	lowScenario	highScenario	topAlternative
C1	Cost	0.1429	0.1286	0.1571	Epsilon
C2	Environmental Management	0.1429	0.1286	0.1571	Epsilon
C3	CO2 Emissions	0.1429	0.1286	0.1571	Epsilon
C4	Labor Practices	0.1429	0.1286	0.1571	Epsilon
C5	Innovation	0.1429	0.1286	0.1571	Epsilon
C6	Community Impact	0.1429	0.1286	0.1571	Epsilon
C7	Waste	0.1429	0.1286	0.1571	Epsilon